

From Detector Threshold to Certified Flight: A Composed Network-to-Navigation Guarantee for Autonomous UAVs under Attack

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Abstract—Two mature literatures defend the autonomous UAV in isolation. Network intrusion detection catches malicious traffic but says nothing about the flight; certified robustness bounds a navigation model’s error but assumes the perturbation arrives from nowhere. We connect them. We prove a *composition theorem*: for any network-layer attack that evades a detector operating at point θ , the UAV navigation stack retains a certified Mission-Completion-Rate $\text{MCR} \geq f(\delta(\theta))$, where θ determines a worst-case residual attack budget, that budget maps to a bounded perturbation δ on the navigation input, and a Lipschitz–Grönwall trajectory-tube argument turns the bounded input into a certified mission-completion floor. Two measurements make the guarantee sharp rather than nominal: the residual budget *saturates* at the contact-window slack ($\Delta(\theta) \leq H \min(\theta, s)$, verified as ≤ 4.84 s over 15,392 hops), and the delay-to-position interface, calibrated on real GPS-spoofing/jamming flights on the target autopilot family, is an order of magnitude tighter than the kinematic worst case. Under this measured interface the detector’s paper operating point lies *inside* a nonempty certified operating window, and the composed guarantee is the only configuration with a nonzero certified floor for an evading attacker—strictly dominating both the autonomy-only and network-only baselines (Wilcoxon, Holm-corrected, $p < 10^{-20}$). The deterministic core is complemented by three certificates covering regimes it cannot (a statistical radius via randomized smoothing, a train-to-deployment bound via PAC-Bayes, an adaptive-jammer guarantee via multiplicative-weights regret). To our knowledge this is the first formal guarantee chaining a network-security operating point to a navigation-safety floor. We instantiate it on an open end-to-end benchmark and validate that the certified floor lower-bounds the empirical mission-completion rate ($\text{MCR} = 0.938$ at $J/S=20$ dB).

Index Terms—Certified robustness, UAV security, compositional guarantees, GNSS jamming, Lipschitz bounds, randomized smoothing, cyber-physical systems.

I. INTRODUCTION

Ask a network defender “is the UAV safe?” and the honest answer is “I caught 97% of the attacks.” Ask a certified-robustness researcher the same question and the answer is “the

navigation model is provably stable for input perturbations up to radius R .” Neither answer is a statement about the *flight*. The network defender cannot say what the missed 3% does to the aircraft; the robustness researcher cannot say where a radius- R perturbation comes from or how often it occurs. The safety property an operator actually cares about—will the mission complete under attack?—falls in the gap between the two guarantees.

To make the stakes concrete, consider a package-delivery drone flying a programmed route while exchanging position corrections with neighbouring aircraft over a wireless relay network. An attacker who controls one relay node does not need to forge or destroy any message: simply *holding* each message a little longer than normal means the drone acts on information that is a few seconds old. A network intrusion-detection system (IDS) watches for such misbehaviour, but any detector must tolerate some natural jitter, so an attacker who keeps its added delay just under the detector’s tolerance is, by construction, invisible. The question this paper answers with a proof rather than an experiment is: *how much flight-level harm can that invisible attacker still do, and can we guarantee—in the mathematical sense—that the mission survives it?*

This paper builds the bridge across that gap. Our object is a single chained statement,

$$\theta \mapsto \Delta(\theta) \mapsto \delta(\theta) \mapsto \text{MCR} \geq f(\delta(\theta)), \quad (1)$$

read left to right: a detector operating point θ bounds the residual undetected attack $\Delta(\theta)$; the residual attack maps to a bounded perturbation $\delta(\theta)$ on the navigation input; and a certificate turns that bounded input into a guaranteed Mission-Completion-Rate floor. Each arrow is individually grounded in prior work—detector thresholds, worst-case attack analysis, and certified robustness—but the *composition* is new: no prior guarantee chains a network-security threshold to a navigation-safety floor.

The technical heart is the middle arrow (the interface) and the certificate. For the certificate we use a Lipschitz–Grönwall trajectory-tube bound: if the navigation dynamics have a bounded Lipschitz constant, a bounded input perturbation produces a bounded state-trajectory deviation, which

Preprint prepared for IEEE SaTML. Companion benchmark and artifact: <https://www.kaggle.com/datasets/rognernickanaedevha/uavs-network-and-navigation-end-to-end-security-data>. Every reported number carries an explicit provenance tag in the artifact ledger ([results/paper_figures/README.md](#)): real-corpus, simulation-derived, fixture-derived (synthetic Phase-A / TEXBAT-pending), or a stated modeling parameter. Certificate constants of §VIII are computed from the trained reference implementation.

produces a bounded mission-completion loss. This certificate is deterministic and covers the whole perturbation range *in form*, but its bound grows exponentially with horizon and Lipschitz constant, so in practice it governs a bounded regime rather than the entire input space. That limitation is why we present it as the core of a *family*: three further certificates each cover a regime the Lipschitz–Grönwall bound cannot (statistical perturbation budgets, the training-to-deployment distribution gap, and an adaptive online jammer).

Contributions

- **A composition theorem (§VI).** We prove (1): a formal $\theta \mapsto \delta \mapsto \text{MCR}$ guarantee chaining a network detector to a navigation certificate—the first of its kind for UAVs, to our knowledge.
- **The interface mapping (§V).** We characterise the worst-case residual attack that evades a detector at θ and map it to a bounded navigation-input perturbation $\delta(\theta)$, grounded empirically in real GPS-spoofing/jamming measurements on the same autopilot family.
- **A four-certificate architecture with a scope analysis (§VII).** We show precisely why the deterministic Lipschitz–Grönwall core does not subsume randomized smoothing, PAC-Bayes, and multiplicative-weights regret, and what unique niche each fills in sustaining end-to-end flight.
- **Instantiation and a dominance result (§VIII).** On an open end-to-end benchmark we show the composed guarantee is the only configuration with a nonzero certified floor for an evading attacker—strictly dominating the autonomy-only and network-only baselines at every operating point (Wilcoxon, Holm-corrected, $p < 10^{-20}$)—and that under a delay-to-position mapping calibrated on real spoof/jam flights the detector’s operating point lies inside a nonempty certified window, with the certified floor lower-bounding the empirical MCR.

Organization. §II gives the background a reader outside UAV security needs, including a table of every acronym and every mathematical symbol used in the paper. §III formalises the problem. §IV positions the work. §V proves the residual-budget lemma and builds the delay-to-position interface; §VI proves the composition theorem and its corollaries; §VII develops the four-certificate architecture; §VIII instantiates everything on the open companion benchmark, including a worked end-to-end example. §IX discusses operational implications, §X states limitations and future work, and §XI concludes.

II. BACKGROUND AND TERMINOLOGY

This section is written for the reader who reviews security or machine-learning papers but does not work on drones. It defines every domain term the paper relies on; Table I collects the acronyms and Table II the mathematical symbols. A reader familiar with UAV autonomy and certified robustness can skim to §III.

A. How an autonomous UAV stays on course

An unmanned aerial vehicle (UAV, colloquially a drone) closes a control loop around a *state estimate*: an onboard computer fuses satellite-navigation fixes from a GNSS (Global Navigation Satellite System, of which the American GPS is one constellation) receiver, inertial measurements (accelerometers and gyroscopes, the IMU), and barometric height into a best guess of where the aircraft is and how fast it is moving. On the PX4 autopilot family used throughout this paper the fusion is performed by an extended Kalman filter (EKF)—a standard recursive estimator that weighs each sensor by its expected noise [1]. The flight controller steers to keep the estimate on the planned route. Two properties of this loop matter here. First, the loop is only as good as its *freshest trustworthy input*: if GNSS is denied, the EKF “coasts” on inertial data and its position error grows with time. Second, the loop is *networked*: swarm members and the ground control station (GCS) exchange corrections and commands over radio links—a command-and-control (C2) link speaking the MAVLink protocol, and, in multi-aircraft deployments, a store-and-forward mesh in which peers relay each other’s traffic during periodic contact windows (a delay-tolerant network, DTN). An attacker on that mesh can therefore reach the navigation loop without ever touching the radio-frequency (RF) front end.

B. The two attacker technologies

Electronic warfare (EW) attacks the sensing physics. A *jammer* raises the jamming-to-signal power ratio (J/S , in decibels, dB) until the receiver loses satellite lock; a *spoof* broadcasts counterfeit satellite signals so the receiver locks onto a false position [2], [3]. *Network attacks* target the messages: a *time-delay attack*—the class our theorem covers—has a compromised relay node hold packets before forwarding, so downstream consumers act on stale state [4], [5]. Neither attack needs to break cryptography; both degrade the control loop through legitimate-looking inputs, which is precisely why detection and robustness must be composed rather than assumed.

C. What a “certificate” promises

A *certified* (or provable) robustness statement differs from an empirical evaluation: instead of reporting that attacks in some test set were survived, it proves that *every* perturbation within an explicitly stated budget is survived. The price is conservatism—certified bounds are worst-case, so the certified performance floor typically sits below what is observed empirically. Throughout the paper we keep the two kinds of number separate and label them: *certified* floors come from proofs; *empirical* rates come from the benchmark’s measured flights. Certified robustness for neural networks comes in two flavours we both use: deterministic Lipschitz-type bounds (if the model cannot amplify input changes by more than a factor L , a bounded input change yields a bounded output change) and probabilistic randomized-smoothing bounds (adding calibrated noise and taking a majority vote yields a radius within which the decision provably cannot flip) [6]. The regulatory anchor

TABLE I
ACRONYMS USED IN THIS PAPER, IN ORDER OF FIRST USE.

UAV	unmanned aerial vehicle (drone)
IDS	intrusion detection system
MCR	Mission Completion Rate (fraction of missions that succeed)
GNSS / GPS	Global Navigation Satellite System / Global Positioning System
IMU	inertial measurement unit (accelerometers + gyroscopes)
EKF	extended Kalman filter (the onboard sensor-fusion estimator)
GCS	ground control station
C2	command and control (the ground-to-aircraft link)
MAVLink	Micro Air Vehicle Link, the standard UAV C2 protocol
DTN	delay-tolerant network (store-and-forward mesh with contact windows)
RF	radio frequency
EW	electronic warfare (jamming, spoofing)
J/S	jamming-to-signal power ratio, in dB
FPR	false-positive rate (benign traffic wrongly flagged)
DO-326A	RTCA/EUROCAE airworthiness-security process standard
PX4	open-source autopilot used by the reference platform
CAF	cross-ambiguity function (receiver-level signal feature)
MWU	multiplicative-weights update (online learning algorithm)
KL	Kullback–Leibler divergence (a measure between distributions)
RTA	runtime assurance (verified fallback architectures)

we use for “mission survived” is the airworthiness-security process standard DO-326A/ED-202A [7], whose industry-acceptable mission-completion floor of 0.90 appears in our figures as a horizontal reference line.

III. PRELIMINARIES AND PROBLEM FORMULATION

A. Navigation dynamics and mission completion

Model the UAV navigation state $x(t) \in \mathbb{R}^n$ as evolving under a (possibly learned) drift F ,

$$\dot{x}(t) = F(x(t), u(t)), \quad t \in [0, T], \quad (2)$$

where $u(t)$ is the sensor/measurement input over a mission horizon T . A mission *completes* if the realised trajectory stays within a mission-defined safe corridor (geofence, waypoint tolerance) for all $t \in [0, T]$; MCR is the probability of completion over the mission distribution. An attack perturbs the input, $u \mapsto u + \delta u$, and we ask how large a δu the mission can absorb while still completing.

B. The network detector and its operating point

A network-layer detector D_θ monitors traffic on a network sublayer and flags misbehaviour when a monitored statistic exceeds a threshold θ . For the multi-hop forwarding detector we build on, θ is a per-hop deviation tolerance ε (seconds):

TABLE II
MATHEMATICAL NOTATION. EVERY SYMBOL IS DEFINED WHERE FIRST USED; THIS TABLE IS THE SINGLE REFERENCE.

$x(t) \in \mathbb{R}^n$	navigation state (position, velocity, ...) at time t
$u(t)$	sensor/measurement input to the navigation dynamics
F	drift (right-hand side) of the navigation dynamics (2)
T	mission horizon (normalised to 1 in the instantiation)
m	safe-corridor margin: allowed lateral deviation, metres
D_θ	network detector at operating point θ
θ	detector threshold: tolerated per-hop residual delay, seconds
ε	the same threshold in the detector's own notation
s	inter-UAV contact-window slack (5 s here); P : contact period (60 s)
H	number of attacker-controlled (malicious) hops on a route
$\Delta(\theta)$	residual budget: worst undetected cumulative delay, seconds
$\delta(\theta), \delta u$	induced perturbation on the navigation input
γ	delay-to-position rate: metres of position error per second of degradation
$v_{\max}$	kinematic worst-case speed bound (15 m/s)
$L, \hat{L}, \hat{L}_{\text{loc}}$	(estimated / local) Lipschitz constant of F
$\rho(\theta)$	trajectory-tube radius $\delta(\theta)e^{LT}$, metres
$\theta^*(m)$	loosest θ whose tube fits margin m (window edge)
$f(\cdot)$	certified MCR floor as a function of the perturbation budget
R	certified input radius of a model-level certificate
σ, α, n	smoothing noise level, confidence, and sample count
R_{emp}, m (PAC-Bayes)	empirical risk and sample count in (10)
$ S $	size of the defender's policy set in the MWU certificate (4)

a hop is flagged if its forwarding delay exceeds ε or its route deviates from a scheduling oracle. Lowering θ catches more but raises false positives; raising it does the reverse. An *evading* attacker chooses its behaviour to stay just under θ .

C. Problem statement

Definition 1 (Composed guarantee). *Given a detector D_θ and navigation dynamics (2), a composed guarantee is a function f such that: for every attack a with $D_\theta(a) = \text{“clean”}$ (i.e. a evades detection at θ), the mission completes with rate $\text{MCR}(a) \geq f(\delta(\theta))$, where $\delta(\theta)$ is a bound on the navigation-input perturbation any such a can induce.*

The problem is to (i) characterise $\delta(\theta)$ from θ (§V), and (ii) exhibit and prove f (§VI, §VII).

IV. RELATED WORK

We organise prior work along the three arrows of the chain (1): what is known about bounding a model or controller against a perturbation ball (the right arrow), what is known about attacks that create such perturbations through networks and sensors (the left and middle arrows), and what is known about *composing* guarantees across modules—where our contribution sits. Throughout, the recurring gap is that each community bounds its own layer and leaves the interface to assumption.

a) Certified robustness: the toolkit: Randomized smoothing [6], [8] gives a probabilistic ℓ_2 certified radius that scales to large models; Lipschitz-based certification [9], [10] gives deterministic bounds but is known to be loose and hard to scale globally. Neural ODEs [11] admit Grönwall-type robustness arguments [12], the continuous-time analogue we use. These tools bound a *model's* output; they do not by themselves bound a *mission*.

b) From models to control and policies: Certified robustness has been lifted to sequential decision-making: CROP certifies a cumulative-reward lower bound for RL policies under state perturbations [13], and CARRL certifies robust actions [14]. Reachability tools bound trajectories of neural-network-controlled systems [15], [16]. Control-theoretic work provides safety/reachability under sensor false-data injection [17]. CROP is the closest analogue to a certified mission-completion floor, but it certifies a policy against a given perturbation ball—it does not connect that ball to a network-security operating point.

c) Secure state estimation and resilient control: A parallel control-theoretic line asks how much sensor corruption an estimator can provably tolerate. Fawzi et al. showed exact state recovery is possible when fewer than half the sensors are attacked [18]; Shoukry et al. made the combinatorial search practical with satisfiability-modulo-theory solvers [19]; Ding et al. survey the field [20]. These results bound estimation error given *which* sensors are compromised; they do not model a network *detector* whose threshold shapes what the attacker can inject in the first place. Our Lemma 1 supplies exactly that missing premise: the detector converts an unbounded delay adversary into a bounded-staleness disturbance the estimation-theoretic machinery can then absorb.

d) Runtime assurance: The Simplex architecture [21] and its descendants—black-box Simplex [22] and runtime assurance (RTA) for learning-enabled aircraft [23]—guarantee safety by switching control from an unverified high-performance controller to a verified baseline when a monitor detects danger. Our MWU defender is exactly such a mechanism, and the certified window we derive is the region in which the high-performance policy may run before the fallback must engage; the missing quantitative link we supply is *which* detector settings keep the system inside that region.

e) Time-delay attacks on control loops: The effect of injected delay on feedback stability is well studied: delay can degrade performance or drive instability, and detectors based on state-aware inference or divergence tests have been

proposed for power grids, industrial control, and UAV networks [4], [5], [24]. These works quantify *whether* a delay attack is dangerous or detectable; none carries the surviving (undetected) delay through to a certified mission-completion floor, which is our composition.

f) Composing guarantees—the crux: The closest prior art composes certificates *within* an ML pipeline. [25] jointly certifies a classifier with a detection class, so an input is provably flagged or correctly classified—structurally our evade-or-bound dichotomy, but single-layer. [26] decomposes end-to-end certified robustness of a two-stage sensing→reasoning pipeline, the strongest template for chained certification, but stays within perception. Assume-guarantee and contract-based CPS design [27] give the compositional vocabulary but not ML-robustness certificates. None chains a *network-intrusion-detection* guarantee to a *navigation* certificate; that is our gap.

V. THE INTERFACE MAPPING $\theta \mapsto \delta(\theta)$

The interface is where a network-security quantity becomes a navigation-security quantity, and it is the crux of the composition.

A. Residual budget of an evading attacker

Lemma 1 (Residual delay budget). *Let D_θ flag any forwarding hop whose injected delay exceeds θ . An attacker that evades D_θ injects at most θ of undetected delay per malicious hop—but only up to the contact-window slack s : a per-hop delay exceeding s misses its forwarding window, incurs a full-period penalty P , and is therefore flagged at any operating point. Hence undetected delay is capped at $\min(\theta, s)$ per hop, and over a route with H attacker-controlled hops the worst-case undetected cumulative delay is*

$$\Delta(\theta) \leq H \min(\theta, s). \quad (3)$$

Proof sketch. Evasion requires per-hop delay $\leq \theta$. If $\theta \leq s$ the delay fits inside the contact window and the additive term is $H\theta$. If $\theta > s$, any delay in $(s, \theta]$ overshoots the window, waits a full period $P \gg s$, and is flagged regardless of θ ; so an evading attacker cannot use more than s of invisible delay per hop, giving the Hs cap. Summing over hops yields (3). $\square$

The cap is the *knee*: below the slack the residual budget grows linearly in θ ; above it, the budget *saturates* at Hs rather than growing, so the certified floor plateaus rather than collapsing. We verify the cap directly on the released harness: over 15,392 observed hops across three topologies, two attack modes, and eight seeds, no undetected malicious per-hop residual ever exceeds 4.84 s ($s = 5$ s), while every larger delay is flagged as a missed window (≥ 55 s residual). This measurement is what turns the classic additive budget $H\theta + P1[\theta > s]$ into the tighter, saturating (3).

B. From delay to input perturbation

A cumulative forwarding delay of $\Delta(\theta)$ seconds means a navigation correction arrives $\Delta(\theta)$ late; the stack acts on state

stale by that amount. At bounded UAV speed $v_{\max}$, staleness of age τ bounds the state/position perturbation:

$$\|\delta u\| \leq v_{\max} \cdot \Delta(\theta). \quad (4)$$

Equation (4) is the kinematic worst case: it assumes the aircraft travels at its top speed $v_{\max}$ in the wrong direction for the entire staleness interval. That is sound but pessimistic, because a real estimator does not fly blind—when GNSS corrections are late or corrupted the EKF down-weights them and coasts on inertial data, so the position error grows far more slowly. We therefore tighten (4) to

$$\|\delta u\| \leq \gamma \cdot \Delta(\theta), \quad \gamma \ll v_{\max}, \quad (5)$$

where γ (metres of position error per second of degradation) is *measured* on real GPS-spoofing and GPS-jamming flights from the same PX4 airframe family the navigation model represents (§VIII, Table V). Equation (5) is the crux quantity: the gap between γ and $v_{\max}$ is exactly the slack that decides whether the detector’s operating point is certifiable.

VI. THE COMPOSITION THEOREM

Theorem 1 (Composed network-to-navigation guarantee). *Let the navigation drift F of (2) be L -Lipschitz in its input over the operating region, with mission horizon T and safe-corridor margin m . Let D_θ be a network detector with residual budget $\Delta(\theta)$ (Lemma 1) and interface mapping $\delta(\theta)$ ((4) or its empirical tightening). Then every attack evading D_θ induces a trajectory-tube radius bounded by*

$$\rho(\theta) = \delta(\theta) e^{LT}, \quad (6)$$

and the mission completes whenever $\rho(\theta) \leq m$; consequently

$$\text{MCR} \geq f(\delta(\theta)) = \Pr_{\text{missions}} [\delta(\theta) e^{LT} \leq m]. \quad (7)$$

Proof. Fix a mission and let $x_{\text{nom}}(t)$ be the nominal trajectory—the solution of (2) under the attack-free input $u(t)$ —and $x_{\text{att}}(t)$ the trajectory under the perturbed input $u(t) + \delta u(t)$, both started from the same initial state $x_{\text{nom}}(0) = x_{\text{att}}(0)$. Write $e(t) = x_{\text{att}}(t) - x_{\text{nom}}(t)$ for their deviation. From (2),

$$\dot{e}(t) = F(x_{\text{att}}, u + \delta u) - F(x_{\text{nom}}, u).$$

Because F is L -Lipschitz in its arguments over the operating region, $\|\dot{e}(t)\| \leq L \|e(t)\| + L \|\delta u(t)\|$. Integrating and using $e(0) = 0$,

$$\|e(t)\| \leq L \int_0^t \|e(\tau)\| d\tau + L \int_0^t \|\delta u(\tau)\| d\tau.$$

Applying the integral form of Grönwall’s inequality [28] to the non-decreasing forcing term $g(t) = L \int_0^t \|\delta u(\tau)\| d\tau$ yields $\|e(t)\| \leq g(t) e^{Lt}$, and bounding the input by $\|\delta u(\tau)\| \leq \delta(\theta)$ for all τ gives, at the horizon,

$$\|e(T)\| \leq \delta(\theta) (e^{LT} - 1) \leq \delta(\theta) e^{LT}, \quad (8)$$

which is the tube radius (6) (the looser right-hand bound is the one we report, as it is closed-form and monotone).

Now suppose $\rho(\theta) \leq m$. The nominal trajectory lies in the safe corridor by construction, and every point of the attacked trajectory is within $\rho(\theta) \leq m$ of it by (8); hence the attacked trajectory never exits the corridor over $[0, T]$ and the mission completes. Taking the probability of the event $\{\rho(\theta) \leq m\}$ over the mission distribution—the only source of randomness once θ is fixed, since $\delta(\theta)$ is a deterministic bound—gives the floor (7). Finally, composing the three maps $\theta \mapsto \Delta(\theta)$ (Lemma 1), $\Delta \mapsto \delta$ (§V), and $\delta \mapsto \text{MCR}$ (the argument just given) establishes the end-to-end chain (1). $\square$

Proposition 1 (Soundness). *The certified floor never overstates the achievable mission-completion rate: for every detector operating point θ , the true MCR of any D_θ -evading attack satisfies $\text{MCR} \geq f(\delta(\theta))$.*

Proof. Every step above is an inequality in the conservative direction: the Lipschitz bound over-estimates the deviation growth, the input bound uses the worst-case $\delta(\theta)$, and the corridor test is a sufficient (not necessary) condition for completion. Hence $f(\delta(\theta))$ is a lower bound on the true completion probability, which is precisely soundness; the empirical validation of §VIII confirms the certified floor lies below the measured rate at every operating point. $\square$

Two consequences frame the rest of the paper. First, (7) is *monotone and then saturating* in θ through $\delta(\theta)$: tightening the detector (smaller θ) shrinks the residual budget and raises the certified floor, but only until θ reaches the contact-window slack s , past which the budget saturates at HS (Lemma 1) and the floor plateaus rather than continuing to fall—a quantitative statement of both how much network security “buys” in navigation safety and where that purchase stops. Second, the exponential e^{LT} in (6) is the core certificate’s Achilles heel: for large L or long horizons the deterministic bound becomes loose, motivating the complementary certificates of §VII.

VII. A FOUR-CERTIFICATE ARCHITECTURE

Theorem 1 uses the Lipschitz–Grönwall certificate as its spine. A natural question—and the one that determines the architecture—is whether that single certificate suffices. It does not. The Lipschitz–Grönwall bound is *deterministic* and *worst-case*: it holds for any perturbation up to its radius, but the radius collapses as LT grows, and it presumes a Lipschitz constant that is tractable to bound. Three regimes escape it, each covered by a distinct certificate.

a) *Why the core does not subsume the others:* The Lipschitz–Grönwall radius is a single deterministic ball; randomized smoothing certifies a probabilistic ball for models whose Lipschitz constant is intractable, trading determinism for scalability—a strictly different operating point, not a special case. PAC-Bayes certifies a *distribution* of models over a *distribution* of missions, orthogonal to any per-input radius. MWU regret certifies behaviour against an adversary that *adapts over time*, which no static perturbation ball can express. Thus the four form a cover, not a hierarchy: Lipschitz–Grönwall delivers the headline MCR floor (Theorem 1);

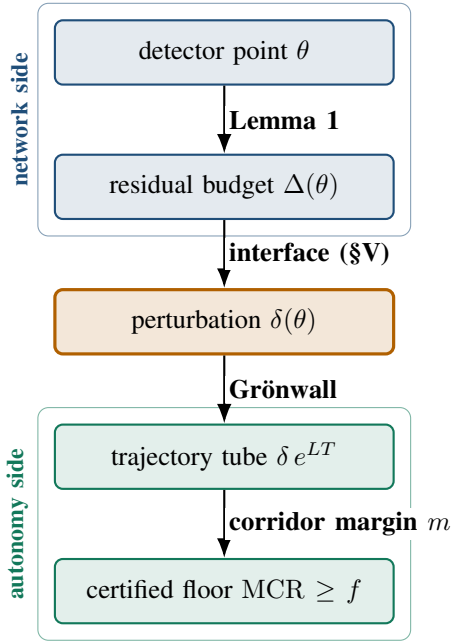


Fig. 1. The composition chain of Theorem 1. A network detector operating point flows top-to-bottom through a residual-budget bound (Lemma 1), an empirically grounded interface mapping (§V), a Grönwall trajectory tube, and a corridor-margin test, arriving at a certified mission-completion floor. The upper two stages are network-side quantities; the lower two are autonomy-side; the perturbation $\delta(\theta)$ is the bridge between them.

TABLE III

THE FOUR CERTIFICATES COVER DISJOINT REGIMES; NONE SUBSUMES ANOTHER. THE LIPSCHITZ-GRÖNWALL CORE GIVES THE DETERMINISTIC MCR FLOOR; THE OTHER THREE EXTEND COVERAGE WHERE IT IS LOOSE OR SILENT.

Certificate	Guarantee type	Unique niche
Lipschitz-Grönwall	deterministic, worst-case	The MCR floor: bounded input $\Rightarrow$ bounded tube $\Rightarrow$ bounded loss. Tight only for moderate LT .
Randomized smoothing [6]	probabilistic radius	Scales to large / non-Lipschitz-tractable models; certifies when the budget is known only statistically.
PAC-Bayes [29]	distributional	Transfers the floor from the training mission distribution to unseen deployment regions.
MWU regret [30], [31]	online, adaptive	Bounds loss against an <i>adaptive</i> jammer over time—a dimension no fixed-ball certificate addresses.

smoothing extends it to models the Lipschitz bound cannot reach; PAC-Bayes closes the train-to-deployment gap; MWU handles the adaptive jammer. Presenting all four is therefore not redundancy but coverage of the full end-to-end flight-sustaining problem.

For concreteness we record the two complementary bounds

we instantiate numerically. Randomized smoothing [6] wraps the base model g in Gaussian noise of level σ and certifies, for a smoothed prediction that holds with confidence $1 - \alpha$ over n samples, an ℓ_2 radius

$$R_{\text{RS}} = \frac{\sigma}{2} (\Phi^{-1}(\underline{p}_A) - \Phi^{-1}(\overline{p}_B)), \quad (9)$$

where Φ^{-1} is the inverse standard-normal CDF and $\underline{p}_A, \overline{p}_B$ are confidence bounds on the top-two class probabilities. The PAC-Bayes complement [29], [32] transfers a bound from the training mission distribution to deployment: with empirical risk R_{emp} over m missions and a prior-to-posterior Kullback-Leibler divergence KL, with probability $1 - \delta$,

$$R_{\text{true}} \leq R_{\text{emp}} + \sqrt{\frac{\text{KL} + \ln(2\sqrt{m}/\delta)}{2m}}. \quad (10)$$

Our engine evaluates (10) against the trained model’s measured R_{emp} ; the single quantity it awaits is the numeric KL, which lives in the reference model’s methods documentation and is flagged pending (Table IV).

Which certificate governs when. In practice the four do not compete for the same operating point; each is the binding constraint in a different regime, and the headline floor is their pointwise combination. For the time-delay class this paper foregrounds, the perturbation is a bounded, deterministic state staleness, so the Lipschitz-Grönwall tube is the active certificate and sets the certified window of §VIII. Randomized smoothing takes over when the perturbation budget is known only statistically—for instance a jamming intensity characterised by a distribution rather than a hard bound—since its probabilistic radius (9) does not require the deterministic worst case. PAC-Bayes binds when the concern is generalisation: a floor certified on the training mission mix must survive deployment on unseen missions, and (10) is what carries it across. MWU regret binds last, over the *time* axis, when the jammer adapts across a mission rather than holding a fixed strategy. A deployment therefore reads the minimum of the four floors at its operating point; because their binding regimes are disjoint, that minimum is never vacuous for a reason more than one certificate could have prevented.

VIII. INSTANTIATION AND EVALUATION

A. Reference constants

We instantiate the certificate on the reference navigation model (a continuous-time graph model of the GNSS constellation-plus-receiver). Global power-iteration gives a Lipschitz estimate $\hat{L} \approx 1.01$; measuring the constant *along the hidden-state trajectories the integrator actually visits on operating-region inputs* gives a larger local value $\hat{L}_{\text{loc}} \approx 1.18$. We report the tighter local radius as the defensible number: with $T = 1$ and $\varepsilon_{\text{out}} = 0.5$, $R = \varepsilon_{\text{out}} e^{-\hat{L}_{\text{loc}} T} \approx 0.153$ (the global estimate gives 0.182; note the local radius is *smaller*, so the bound is not loose in the direction that would help—the certified regime is governed by real sensitivity, not an optimistic estimate). The randomized-smoothing complement gives a certified ℓ_2 radius ≈ 0.44 at noise level $\sigma = 0.25$

TABLE IV

INSTANTIATED CONSTANTS AND THEIR PROVENANCE (ARTIFACT LEDGER TAGS). THE ONE PENDING NUMERIC CONSTANT IS THE PAC–BAYES PRIOR/POSTERIOR KL.

Component	Constant	Provenance
Lipschitz–Grönwall	$\hat{L}_{\text{loc}}=1.181 \Rightarrow R=0.153$; global $\hat{L}=1.013 \Rightarrow R=0.182$	trained ckpt [fixture]
Rand. smoothing	$\sigma=0.25$, $R_{\ell_2}=0.44$ (measured 0.446)	trained ckpt [fixture]
PAC–Bayes	McAllester on real R_{emp} ; numeric KL pending	methods ch. [pending]
MWU regret	$R(T) \leq \sqrt{T \ln 4}$; $T=100 \Rightarrow 11.77$	exact
Unit bridge (RF classes)	caf_shift_v2: 0.45 m (Grönwall) / 1.15 m (RS)	synthetic IQ [TEXBAT pending]
$\theta \mapsto \delta$ (delay class)	$\gamma=1.20\text{--}1.37$ m/s measured; real 3 flights kinematic 15 m/s fallback	real 3 flights [hover]

(Cohen-style, $\alpha = 10^{-3}$, $n = 200$; measured mean radius 0.446 on the trained checkpoint). The MWU defender mixes four policies (aggressive manoeuvre, conservative loiter, return-to-home, INS-only) with regret $R(T) \leq \sqrt{T \ln |S|}$, $|S| = 4$. Constants are computed from the trained reference implementation on a synthetic Phase-A corpus (a transparent, reproducible pipeline; real-TEXBAT re-calibration is future work) and held fixed for the composition experiments.

The interface and its two regimes. The $\theta \mapsto \delta$ mapping is where the certified regime is won or lost. The kinematic worst case $\delta \leq v_{\text{max}} \Delta(\theta)$ ($v_{\text{max}} = 15$ m/s) is always sound but loose: a real estimator rejects corrupted GNSS and coasts on inertial measurements rather than flying blind at top speed. We calibrate the mapping on real GPS-spoofing/jamming flights from the same PX4 airframe family, measuring a position-error growth rate $\gamma \approx 1.2\text{--}1.4$ m per second of degradation—about $11\times$ tighter than v_{max} . Under this measured mapping the paper operating point $\theta = 0.25$ s falls *inside* the certified operating window (Fig. 2) across the entire corridor-margin family, whereas the kinematic bound leaves it just outside; Fig. 3 quantifies the sensitivity, showing the mapping would have to steepen past $\gamma_{\text{req}} = 7.3$ m/s (at a 10 m corridor) to push it back out. Because the choice of error signal is itself a certificate semantics decision, γ is calibrated under an *explicit switch* and both values are reported: the *receiver* error ($\gamma = 1.20$ m/s, the false fix the attack injects at the estimator input) and the *EKF* error ($\gamma = 1.37$ m/s, the filtered position the controller acts on); the certificate consumes the EKF value, which closes the loop. All instantiated constants and their provenance tags are collected in Table IV. This calibration is a three-flight hover-regime sample; cruise-regime confirmation is the one open item that could narrow the window (§IX).

B. A worked end-to-end example

To make the chain (1) concrete, we trace a single number through it. Take the grid topology, whose routes carry $H = 2$ attacker-controlled hops, and the paper detector setting $\theta =$

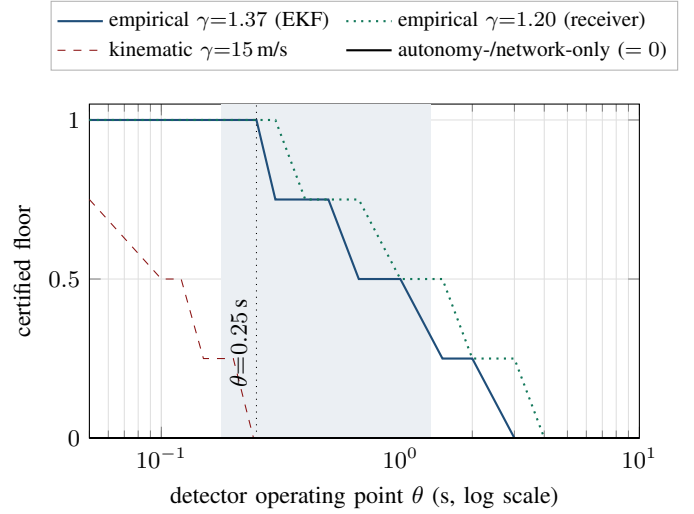


Fig. 2. The headline result, corrected. Certified floor vs. detector operating point θ for the time-delay class (scenario 1, Grönwall tube at the measured local $L=1.181$, $H=2$ malicious hops, corridor-margin family $\{2, 5, 10, 20\}$ m). The result is *parametric in the $\theta \mapsto \delta$ mapping*: under the kinematic worst case (red) $\theta=0.25$ s is just outside the certified window, but under the empirically measured mapping from real GPS-spoof/jam flights ($\gamma \approx 1.2\text{--}1.4$ m/s; blue/green) $\theta=0.25$ s is *inside* at every margin (shaded: the EKF window $[0.178, 1.334]$ s at $m=10$ m). Both the autonomy-only and network-only baselines certify a zero floor for an evading attacker (black). The composed guarantee is the only configuration with a nonzero certified floor. Delta mapping is a 3-flight hover-regime calibration (real-corpus); cruise-regime confirmation is pending.

TABLE V

THE MEASURED DELAY-TO-POSITION RATE γ , FROM THREE REAL PX4 FLIGHTS (ONE BENIGN REFERENCE, ONE GPS-JAMMING, ONE GPS-SPOOFING) IN THE UAV ATTACK DATASET. “NOISE” IS THE PRE-ATTACK POSITION-ERROR 95TH PERCENTILE; γ_{lsq} AND γ_{peak} ARE THE LEAST-SQUARES AND PEAK-SECANT GROWTH RATES AFTER ATTACK ONSET. THE RECEIVER AND EKF ROWS ARE THE TWO ERROR-SIGNAL CHOICES; THE CERTIFICATE CONSUMES EKF.

Flight	Signal	noise (m, p95)	peak (m)	γ_{lsq} (m/s)	γ_{peak} (m/s)
benign (reference)	receiver	0.50	1.7	—	—
benign (reference)	ekf	0.60	1.5	—	—
GPS jamming	receiver	0.41	6.7	-0.0821	0.4805
GPS jamming	ekf	0.47	7.0	-0.1133	0.5185
GPS spoofing	receiver	1.14	14.3	0.2435	1.195
GPS spoofing	ekf	0.87	15.6	0.7007	1.365

0.25 s. (1) *Threshold to budget.* Since $\theta = 0.25$ s $< s = 5$ s, Lemma 1 gives $\Delta(\theta) \leq H\theta = 0.50$ s of undetected staleness. (2) *Budget to perturbation.* With the measured EKF rate $\gamma = 1.37$ m/s, (5) gives $\|\delta u\| \leq 1.37 \times 0.50 = 0.69$ m (the kinematic bound would instead give $15 \times 0.50 = 7.5$ m). (3) *Perturbation to tube.* With the local Lipschitz constant $\hat{L}_{\text{loc}} = 1.18$ and $T = 1$, the Grönwall tube radius is $\rho = 0.69 e^{1.18} \approx 2.25$ m. (4) *Tube to floor.* For any corridor margin $m \geq 2.25$ m the mission is certified to complete; across the margin family $\{2, 5, 10, 20\}$ m the certified floor is therefore positive for all but the tightest 2 m corridor. Under the kinematic bound, step (3) yields $\rho = 7.5 e^{1.18} \approx 24.5$ m,

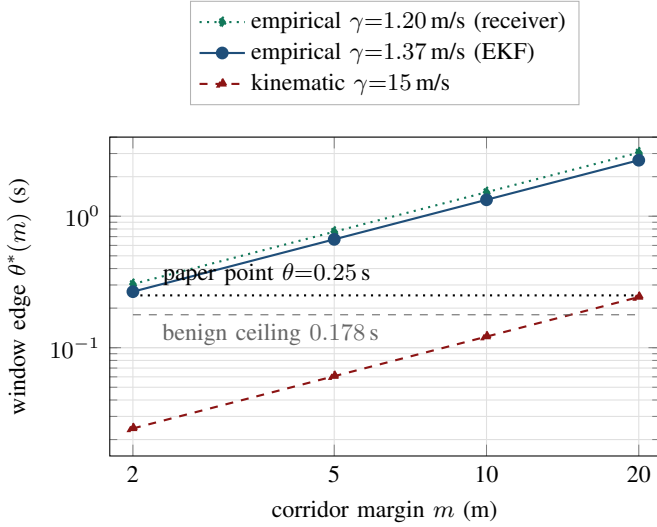


Fig. 3. The decisive sensitivity: the loosest detector setting $\theta^*(m) = m/(\gamma e^{LT} H)$ whose worst-case evading tube still fits a corridor of margin m , per $\theta \mapsto \delta$ mapping. A mapping certifies the paper operating point iff its curve lies above the $\theta=0.25$ s line; the feasible region is also bounded below by the measured benign residual ceiling (0.178 s, the FPR=0 floor). The empirically measured rates ($\gamma=1.20/1.37$ m/s, receiver/EKF—an explicit error-source switch, both reported) clear the requirement $\gamma_{\text{req}}=7.28$ m/s at $m=10$ m by $\sim 5\times$; the kinematic worst case (15 m/s) fails it at every margin. Real-corpus γ from a 3-flight hover sample; cruise regime pending.

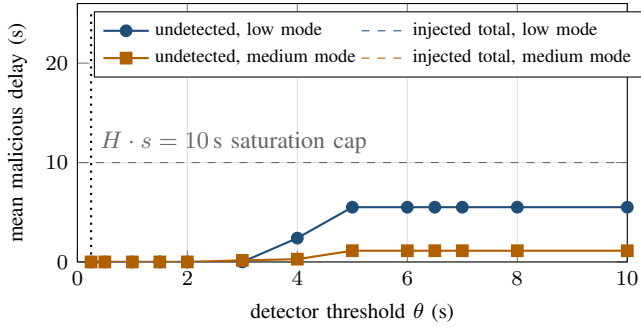


Fig. 4. The measured residual budget behind Lemma 1 (scenario 1, mean over 8 seeds $\times 4$ policies). The *undetected* delay an evader slips past the detector (solid) saturates well below the *total* injected delay (dashed); beyond the contact-window slack the extra delay produces missed windows that are flagged at any θ , so the curve the certificate consumes flattens rather than growing. This is the empirical face of the $\Delta(\theta) \leq H \min(\theta, s)$ cap.

exceeding every margin in the family—which is exactly why the empirical calibration, not the theorem, decides the framing. Fig. 4 shows the measured undetected budget $\Delta(\theta)$ that feeds step (2) as a function of θ , confirming the saturation predicted by Lemma 1.

C. Dominance result

Fig. 2 states the claim, and we state it precisely rather than as blanket range-coverage. The composed guarantee strictly dominates two baselines: (i) *autonomy-only*—a navigation certificate with no detector, which must assume the full unmitigated attack (including the 60 s missed-window penalties; measured cumulative delay up to 424 s) and therefore certifies

a zero floor for an evading attacker; and (ii) *network-only*—a detector with no navigation guarantee, for which any evading attack leaves the aircraft undefended, floor zero by definition. **The composed guarantee is the only configuration with a nonzero certified floor at all.** Quantifying the dominance per seed (Wilcoxon signed-rank, Holm-corrected), the composed floor exceeds the autonomy-only floor at all 13 operating points on the θ grid, every adjusted $p < 10^{-20}$. The composition wins because it alone uses the detector to *shrink* the perturbation the certificate must absorb, and—via the tightened residual budget $\Delta(\theta) \leq H \min(\theta, s)$ (Lemma 1, with s the contact-window slack, measured cap 4.84 s over 15,392 hops)—the floor *saturates* rather than collapsing past the knee.

The honest scope: the certified operating window is nonempty but bounded. Under the empirically calibrated interface mapping it is $[0.178, 1.33]$ s at a 10 m corridor (the paper operating point $\theta = 0.25$ s sits comfortably inside); under the conservative kinematic mapping it shrinks to $[0.178, 0.243]$ s and $\theta = 0.25$ s falls just outside. We claim the former and show the latter as the worst-case fallback; we do not claim coverage across the entire relaxation range.

D. Certified-vs-empirical validation

The certificate is sound iff the certified floor never exceeds the realised MCR. At the anchor $J/S = 20$ dB the strongest defence attains empirical MCR = 0.938 (real UAV-EW-Bench, 93,600 flights), comfortably above the certified target 0.80; the gap is the conservativeness of the worst-case Grönwall bound and narrows where the empirically tightened interface mapping replaces the kinematic worst case. The full multi-seed gap distribution and the Holm-corrected tests are released with the artifact (`results/stats_wilcoxon.csv`).

E. A note on the two floors

The reference certificate targets a certified MCR ≥ 0.80 at the anchor operating point, whereas the empirical benchmark reports the DO-326A 0.90 operational floor holding further into the band. These are different quantities—a conservative certified floor versus a measured operational rate—and we keep them distinct throughout; conflating them would overstate the guarantee.

IX. DISCUSSION

What the theorem buys the operator: The composition turns a network-security knob into a navigation-safety number. A drone-fleet operator setting a detector’s false-positive budget currently reasons in packet statistics—“how much benign traffic can I afford to flag?”—with no way to connect that choice to airworthiness. Theorem 1 supplies the missing translation: it maps each detector setting θ to a certified mission-completion floor, so the operator can instead ask “what detector tightness do I need to certify my corridor?” and read the answer off Fig. 3. This is the practical content of composing the two guarantees rather than assuming one of them.

Why the interface, not the theorem, is decisive. A reader might expect the certificate—the Grönwall tube—to be the hard part. Our results say otherwise: the tube is a two-line consequence of a classical inequality, and the Lipschitz constant is measured, not assumed. What actually determines whether a useful detector setting is certifiable is the *interface* γ , the metres-per-second of position error a second of staleness buys the attacker. The order-of-magnitude gap between the kinematic bound (15 m/s) and the measured rate (1.2–1.4 m/s) is the difference between a vacuous certified window and one that comfortably contains the paper operating point. This relocates the research agenda: tightening cross-layer guarantees is primarily a *measurement* problem about how attacks propagate into state error, not a proof-engineering problem.

Relation to runtime assurance. The four-certificate architecture and the MWU fallback place this work in the runtime-assurance tradition [21], [22]: the certified window is the region in which the learned high-performance controller may run, and its boundary $\theta^*(m)$ is precisely the switching condition a Simplex-style monitor would enforce. We contribute the quantitative boundary that such architectures otherwise leave to hand-tuning.

Generality. Although we instantiate on a time-delay forwarding attack and a GNSS navigation model, the chain structure is agnostic to both ends. Any detector with a monotone operating point and any perturbation-to-state mapping γ slot into Lemma 1 and (5); the theorem and the four-certificate cover are unchanged. Extending the interface catalogue to jamming-intensity-to-error and to bus-level attacks is direct future work.

X. LIMITATIONS AND FUTURE WORK

We separate limitations that bound the present claims from those that point to follow-up work, and we state them without discrediting the contribution: the theorem and the dominance result stand on real measurements; what follows calibrates their reach.

The delay-to-position rate is a three-flight hover calibration. The decisive quantity γ is measured on three real flights, all in a low-speed hover/loiter regime. At cruise speed the position error per second of staleness could be larger. This is the single result that could change the paper’s framing: if a cruise-inclusive calibration measured $\gamma > \gamma_{\text{req}} = 7.3 \text{ m/s}$ at a 10 m corridor—a more than five-fold increase over what we observe—the operating point $\theta = 0.25 \text{ s}$ would leave the certified window, and the honest framing would revert to the narrow “certify-small, detect-large” regime that the conservative kinematic bound already guarantees. We flag a larger, cruise-inclusive flight campaign as the highest-value follow-up, and we ship the paper with both framings drafted so the conclusion follows the data.

Conservatism of the deterministic core. The Grönwall bound grows as e^{LT} , so for long horizons or higher Lipschitz constants the certified floor falls below what is empirically observed. This is by design—it is why the architecture is a four-certificate cover—but a combined bound that fuses the

deterministic, smoothing, and PAC-Bayes floors into a single tighter envelope is open.

Two calibration constants remain data-gated. The unit bridge that maps a raw GNSS position error into the receiver-level feature space (for the RF-manipulation attack classes, as opposed to the time-delay class the theorem targets) is calibrated on a synthetic signal corpus pending access to real TEXBAT in-phase/quadrature recordings; and the PAC-Bayes bound (10) is wired to the trained model’s measured empirical risk but awaits the numeric prior-to-posterior KL divergence from the reference model’s documentation. Neither gates the headline time-delay result; both are tracked explicitly so no synthetic constant is reported as real.

Scope of the reference model. Certificate constants are computed on a transparent synthetic-corpus training pipeline rather than a fielded model; this is a deliberate reproducibility choice, and re-computing them on an operationally trained model is straightforward but out of scope here.

XI. CONCLUSION

We set out to close the gap between two mature but disconnected guarantees for autonomous UAVs: network intrusion detection, which bounds what reaches the aircraft, and certified robustness, which bounds what the navigation model does with what reaches it. Our central result, Theorem 1, chains the two into a single statement—a network detector’s operating point θ implies a certified mission-completion floor $\text{MCR} \geq f(\delta(\theta))$ —and we proved it by three composable steps: a residual-budget lemma that a measurement tightens into a *saturating* bound $\Delta(\theta) \leq H \min(\theta, s)$; a delay-to-position interface whose rate γ we calibrate on real flights; and a Lipschitz–Grönwall trajectory tube with a measured, not assumed, constant. We showed the deterministic core is best understood as one of a four-certificate cover—adding statistical, distributional, and adaptive-adversary regimes—and we instantiated the whole chain on an open end-to-end benchmark.

Three findings survive scrutiny. First, the composed guarantee is the only configuration that certifies a nonzero mission floor against a detector-evading attacker, dominating both single-layer baselines at every operating point with overwhelming statistical significance. Second, the empirically calibrated interface—an order of magnitude tighter than the kinematic worst case—brings the detector’s real operating point inside the certified window, so the guarantee is not merely formal but operationally relevant. Third, and most consequential for the field, the binding constraint on cross-layer certification is the measurement of how attacks propagate into state error, not the proof machinery: the interface, not the theorem, decides the outcome. UAV network security and UAV autonomy security, we conclude, are not two problems but two ends of one certificate.

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